Knowledge & Reasoning - First Order Logic

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Outline

- Why FOL?
- Syntax and semantics of FOL

Logic

- Logic is one of the essential ideas in the field of artificial intelligence.
- AI systems' reasoning and decision-making are based on logic.
- It offers a method for presenting and handling data in a way that enables an AI system to draw conclusions and inferences from it.

$$\underbrace{i^2 + 3k}_{\text{Subject}} \underbrace{\geq 10 + j}_{\text{Predicate}}$$

First-order logic

It allows us to express statements about **objects** and their **relationships** in a precise and unambiguous way.

- Example
  - All men are mortal.
  - Some birds can't fly.
  - At least one planet has life on it.

Ex 1 : Parrots are green.

Parrots(X)→Green(X)

Ex 2: All kids like toys.

like (kids , toys).

Ex 3: Some kids like toys.

like (kids , toys).

First-order logic (continued)

- First-order logic (like natural language) assumes the world contains
  - Objects:
    - people, houses, numbers, colors, baseball games, wars, …
  - Relations:
    - red, round, prime, brother of, bigger than, part of, comes between, …
  - Functions:
    - father of, best friend, one more than, plus, …

Syntax of FOL

- The syntax of First-Order Logic can be defined in terms of
  - Terms
  - Predicates
  - Quantifiers

Term

- A term denotes some object other than true or false.

Term

Tommy is a dog

All men are mortal

Terms: Constant & Variable

- A constant of type W is a name that denotes a particular object in a set W.
  - Example: 5, Tommy etc.

- A variable of type W is a name that can denote any element in the set W
  - Examples: x belongs N denotes a natural number, d denotes the name of a dog

Terms: Functions

- A functional terms of arity n takes n objects of type W_1 to W_n as inputs and returns an object of type W.
- i.e. functional relations.

$$f(W_1, W_2, \ldots, W_n)$$

$$plus(3,4)=7$$

functional term constant term

Functions: Example

- Let plus be a function that takes two arguments of type natural number and returns a natural number
- Valid functional terms:

plus(2,3) plus(5,plus(7,3))

plus(plus(100, plus(1,6)),plus(3,3))

- Invalid functional terms:

plus(0,-1) plus(1.2,3.1)

Predicates

- Predicates are like functions except that their return type is true/false. i.e. relations
- Example
  - gt(x,y) is true iff x>y
  - Here gt is a predicate symbol that takes two arguments of type natural number
  - gt(3,4) is valid predicate
  - but gt(3,-4) is invalid predicate.

Types of Predicates

- A predicate with no variable is a proposition
  - Tommy is a dog
- A predicate with one variable is called a property
  - dog(x) is true iff x is a dog.
  - Mortal(y) is true iff y is mortal.

Formulation of Predicates

- Let P(x,y,…) and Q(x,y…) are two predicates.
- Then

$P \wedge Q$

$P \vee Q$

$\neg P$

$P \rightarrow Q$

$P \Leftrightarrow Q$

Formulation of Predicates (continued)

- **Truth Table for Logical Connectives**

P	Q	¬P	P ∧ Q	P ∨ Q	P -> Q
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	F	T	T
F	F	T	F	F	T

Predicate Examples

- If x is a man then x is mortal

  man(x) → mortal(x)

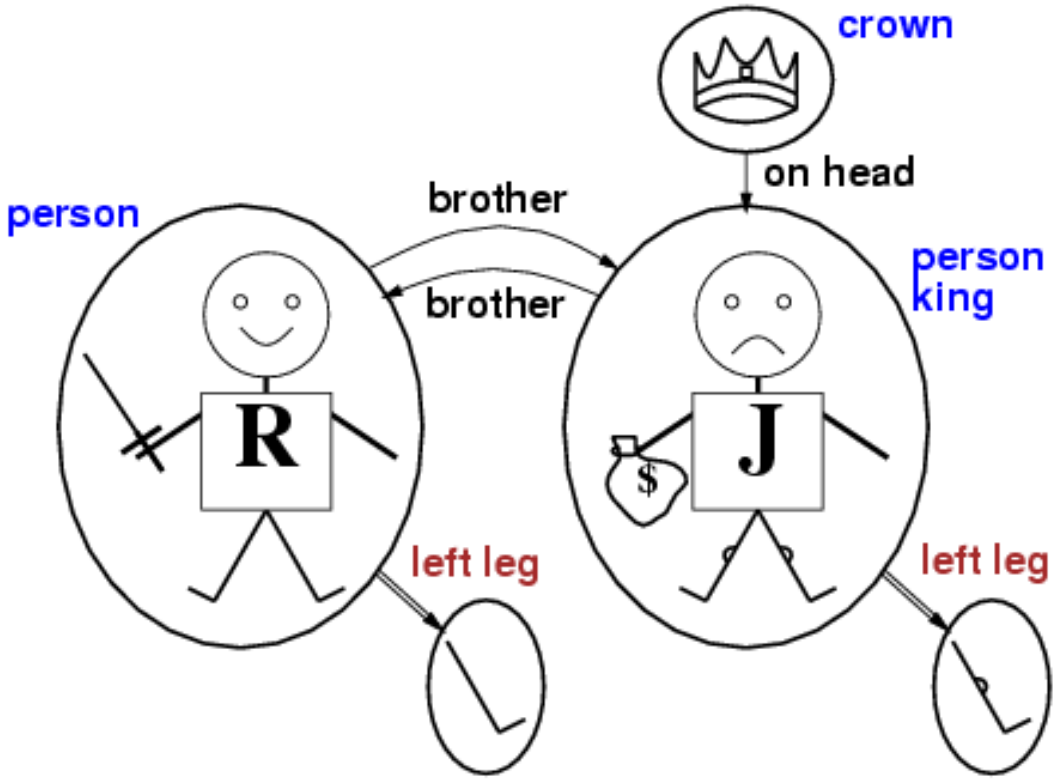
  equivalent: ¬ man(x) V mortal(x)

- If n is a natural number, then n is either even or odd.

  natural(n) →even(n) V odd(n)

Models for FOL: Example

- The domain of a model is the set of objects it contains: five objects.
- Relation in this: Richard (R) refers Richard the Lion Heart, John refers to Evil King John (J), Brother refers to brotherhood relation etc.



- Example
  - Brother(Richard, John)
  - King(Richard) V King(John)

Quantifiers

- Allow to make statements about entire collections of objects
- There are two basic quantifiers in FOL
  - Universal Quantifiers : make statements about everything
    - ∀ “ For all”
  - Existential Quantifiers : make statements about some things
    - ∃ “ There exists”

Universal Quantifiers

- All kings are persons.

  king(x) : x is king

  person(x) : x is person

  $\forall x$ (king(x) -> person(x))

Universal Quantifiers (continued)

- All dogs are faithful

  faithful(x) : x is faithful

  dog(x) : x is a dog

  $\forall x$ (dog(x) -> faithful(x))

Universal Quantifiers (continued)

- All birds can’t fly

  fly(x) : x can fly

  bird(x) : x is a bird

  ¬(∀x (bird(x) -> fly(x)))

Universal Quantifiers (continued)

- Everyone studies in Berkeley is smart

  smart(x) : x is smart

  studiesAt(x, Berkeley) : x studies in Berkeley

  ∀x (studiesAt(x, Berkeley) -> smart(x))

Existential Quantifiers

- All birds can't fly ⇔ There exists a bird that can't fly

  fly(x) : x can fly

  bird(x) : x is a bird

  $\exists x\ (bird(x) \land \neg fly(x))$

Sentences

- A predicate is a sentence
- If sen, sen’ are sentences and x is a variable then

  (sen), ¬sen, ∃ x sen, ∀x sen,

  sen ^ sen’, sen V sen’, sen -> sen’

  are sentences
- Nothing else is a sentence

Examples of Sentences

birthday(x,y)

- x celebrates birthday on date y.

∀y, ∃ x birthday(x,y)

- For all dates, there exists a person who celebrates his/her birthday on that date.
- i.e. everyday someone celebrates his/her birthday.

Express sentence into FOL

- Everybody is intelligent.

  intelligent(x) : x is intelligent

  $\forall x$ intelligent(x)
- Somebody is intelligent.

  $\exists$ x intelligent(x)
- All Students are intelligent.

  student(x) : x is a student

  $\forall x$ student(x) -> intelligent(x)

Express sentence into FOL (continued)

- Everybody likes Icecream.

  $\forall x$ likes(x, Icecream)

- Cats are mammals.

  $\forall x$ cat(x) -> mammal(x)

Express sentence into FOL (continued)

- Everybody loves somebody.

  $\forall x, \exists y \text{ loves}(x,y)$

for all x, there exists a y such that loves(x,y).

So, for each person, there is someone that they love.

$\exists y, \forall x \text{ loves}(x,y)$

There exists a y such that for all x, loves(x,y).

Interpretation is that there is a particular person that everybody loves/ there is someone who is loved by everyone.

Exercise

1. All birds fly.

$\forall x\ bird(x) \rightarrow fly(x)$.

2. Every man respects his parent.

$\forall x\ man(x) \rightarrow respects(x, parent)$.

3. Some boys play cricket.

$\exists x\ boys(x) \wedge play(x, cricket)$.

4. Bill is a student.

$Student(Bill)$

5. Bill takes either Analysis or Geometry (but not both)

$Takes(Bill, Analysis) \Leftrightarrow \neg Takes(Bill, Geometry)$

Duality of Quantifiers

- The two quantifiers are actually intimately connected with each other, through negation.
- Example – Everyone likes ice cream.

  $\forall x$ Likes(x, Icecream)
- There is no one who does not like ice cream.

  $\neg\exists$ x, $\neg$Likes(x, Icecream)

De Morgan's Rule

- ∀ is a conjunction over the universe of object and ∃ is a disjunction.

$$\forall x\ \neg P \equiv \neg \exists\ x\ P$$
$$\neg \forall x\ P \equiv \exists\ x\ \neg P$$
$$\forall x\ P \equiv \neg \exists\ x\ \neg P$$
$$\exists x\ P \equiv \neg \forall x\ \neg P$$

$$\neg P \wedge \neg Q \equiv \neg(P \vee Q)$$
$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$
$$P \wedge Q \equiv \neg(\neg P \vee \neg Q)$$
$$P \vee Q \equiv \neg(\neg P \wedge \neg Q)$$

Equality

- In FOL with equality, we are allowed to use the equality (=) between two functions.
- **$term_1$ = $term_2$** is true under a given interpretation if and only if $term_1$ and $term_2$ refer to the same object.
  - Can be used to state facts about a given function
    - E.g. father(John) = Henry
  - Can ∃ be used with negation to insist that two terms are not the same object.
    - E.g. richard has at least two brothers

∃ x, y brothrer(x, Richard) ∧ brother(y, Richard) ∧ ¬(x=y)

Exercise

- Some dogs bark.

  $\exists x(dog(x) \wedge bark(x))$
- All dogs have four legs.

  $\forall x(dog(x) \rightarrow have_four_legs(x))$

  $\forall x(dog(x) \rightarrow legs(x,4))$
- All barking dogs are irritating.

  $\forall x((dog(x) \wedge bark(x)) \rightarrow irritaging(x))$

Exercise

- Fathers are male parents with children.

$\forall x\ (father(x) \rightarrow male(x) \wedge has_children(x))$

- Spot has atleast two sisters.

$\exists\ x, y\ sisters(x, spot) \wedge sister(y, spot) \wedge \neg(x=y)$

Syntax of FOL: Basic elements

- Constants KingJohn, 2, NUS,...
- Predicates Brother, >,...
- Functions Sqrt, LeftLegOf,...
- Variables x, y, a, b,...
- Connectives ¬, ⇒, ∧, ∨, ⇔
- Equality =
- Quantifiers ∀, ∃

Question ?